

AERO 689 Project Instructions

Aerospace Engineering, Texas A&M University

Instructor: Raktim Bhattacharya

1. Problem Selection

Students should choose a project problem that is:

- Moderately complex: more than a simple tutorial but small enough to scope in 2 weeks.
- Aerospace engineering relevant: examples include aircraft performance, flight systems, control, structural health monitoring, spacecraft operations, launch vehicles, propulsion, aerodynamics, or autonomy.
- Feasible with available data and tools: a problem that can be framed as a prediction, classification, anomaly detection, optimization, or physics-informed ML task.

Some examples:

- Predicting aerodynamic coefficients from limited flight or CFD data.
- Detecting structural damage or fatigue from sensor data.
- Classifying flight phases or operational modes using flight data.
- Estimating vehicle state or trajectory with a hybrid physics+ML model.
- Identifying anomalies in propulsion or thermal systems.

These are just examples. Students may work on any aerospace-relevant problem they are interested in, as long as it is scoped for roughly two weeks.

Avoid overly broad topics or full-scale system development that would take much longer than two weeks.

2. Preliminary Submission Deadline

- **Preliminary work due:** April 19, 12:00 PM Central US Time.
- This submission should include your problem statement, project plan, and the first-report draft as described below.

3. First Report Guidance

The first report should read like an NSF-style white paper or proposal. It should be concise, professional, and focused on the engineering case for the project. This report is also training for writing grant-style submissions and technical reports that are expected in aerospace careers and graduate work.

Required sections

1. **Title and Team**
 - Project title.

- Team members and roles.
- 2. **Problem Statement**
 - What aerospace problem are you solving?
 - Why is it important for aerospace engineering or operations?
 - What are the engineering consequences if the problem is not solved?
- 3. **Background and Motivation**
 - Brief context and relevant existing work.
 - Why is the problem challenging today?
 - What gap does this project address?
- 4. **Proposed Approach**
 - What machine learning or hybrid modeling method will you use?
 - What are the model inputs and outputs?
 - Why is this approach appropriate for the problem?
- 5. **Data and Methods**
 - What data sources will you use?
 - How will you collect, preprocess, and validate the data?
 - What performance metrics will you use to evaluate success?
- 6. **Expected Outcomes**
 - What do you expect to deliver at the end of the project?
 - How will you know if the project succeeded?
- 7. **Timeline and Milestones**
 - A simple two-week plan with milestones.
 - Include at least one intermediate checkpoint before the final submission.
- 8. **Risks and Mitigation**
 - What are the main risks or challenges?
 - How will you mitigate them?
- 9. **Aerospace Impact**
 - Why does this matter for aerospace engineering specifically?
 - How will the result translate to aerospace applications?

4. Rubric for the First Report

The report will be evaluated on the following criteria:

- **Problem clarity and relevance (25%)**
 - The aerospace problem is clearly defined.
 - The motivation is strong and relevant.
- **Technical plan and feasibility (25%)**
 - The proposed approach is appropriate and realistic for a 2-week effort.
 - The plan includes a coherent ML or hybrid modeling strategy.
- **Data and evaluation (20%)**
 - The report identifies usable data sources and preprocessing needs.
 - Performance metrics are meaningful and aligned with the objective.
- **Timeline and risk management (15%)**
 - A feasible schedule is presented.

- Key risks are identified with mitigation strategies.
- **Writing quality and presentation (15%)**
 - The report is organized, professional, and concise.
 - The writing demonstrates clear communication, strong technical structure, and a grant-style narrative.
 - Figures, tables, or diagrams (if used) support the narrative.

5. Submission Notes

- Treat this report as a technical white paper: concise, clearly written, and focused on engineering impact.
- Use this assignment as practice in writing grant-style proposals and technical reports, with clear problem framing, logical flow, and professional presentation.
- Students may use LLM tools to help draft text, brainstorm structure, or check grammar, but must verify all technical content and cite or acknowledge any AI assistance. The instructor will check the technical correctness of the work and may ask for clarification if AI-assisted content is used.
- Keep the document short (2-4 pages is ideal) but complete enough for the instructor to understand the project scope and plan.
- Include references to any prior work, datasets, or algorithms that inspired the approach.